

Assignment Sheet

Task 1 – Pseudoinstructions

(3 Points)

What Instruction Format do pseudo-instructions use?

Task 2 – LOAD-STORE Architecture

(3 Points)

Explain in your own words what a LOAD-STORE Architecture is.

Task 3 – Shifts

(4+4+4 Points)

- a) The following code is intended to perform an arithmetic shift right by 5 bits but gives the wrong result for negative numbers:

```
srl t0, t0, 5
```

Identify the error and rewrite the instruction correctly.

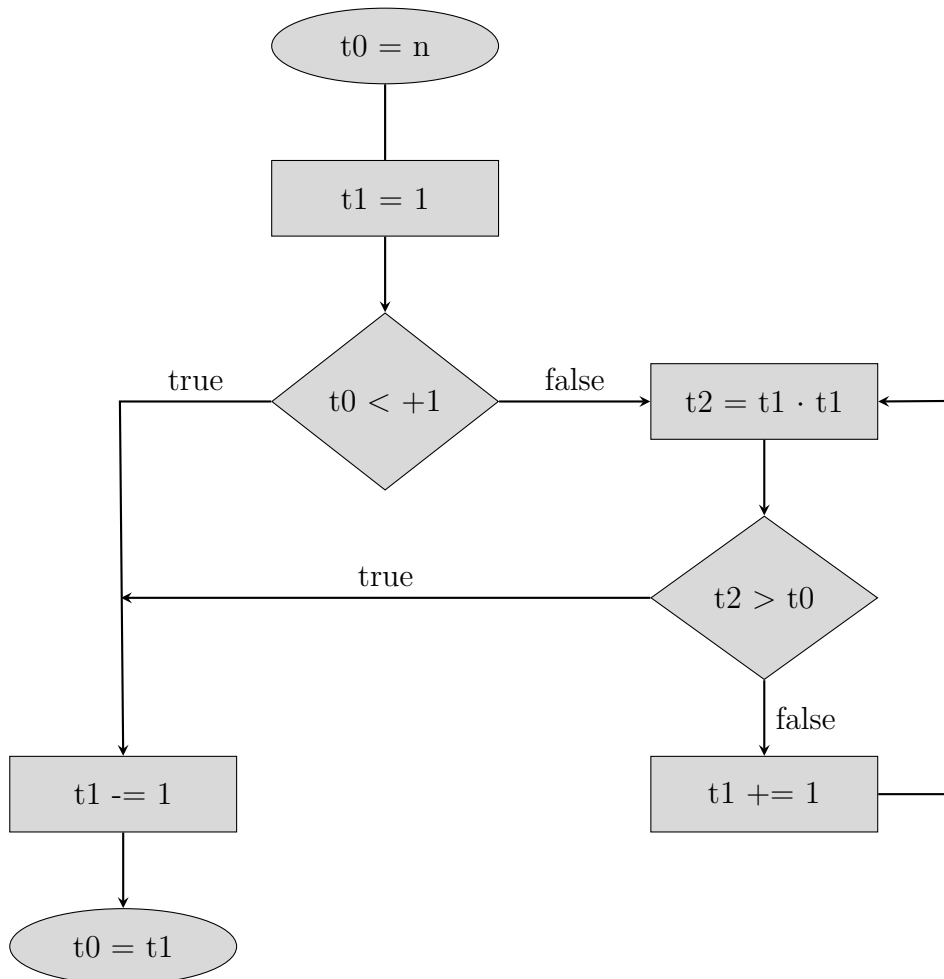
- b) In your own words, explain the difference between sra and srl. Why do we need both instructions?

- c) What instruction can you use to divide an unsigned number given in register t0 by 16 using only shifts?

Task 4 – Flow chart

(2+10 Points)

The following flowchart calculates a mathematical function:



a) What function is described by the flow chart?